Grasp to Identify: Physical-Only Proprioceptive Recognition of Fifteen Classes with a LEAP Hand

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Abstract

Robotic hands commonly use vision or dedicated tactile sensors to recognise objects. Using 15 household objects, this study tests whether a physical LEAP Hand can distinguish among objects based solely on the joint angles achieved during controlled grasps. Motor-current thresholds stop an entire finger when resistance is detected. Three sequential grips on the object provide 48 final angles per trial to a 24-unit hidden-layer classifier. The dataset contains five physical trials for each of 15 labels, comprising 14 objects and an empty-hand condition. In three-fold within-dataset validation, the neural network correctly classified 70 of 75 held-out trials. The decision rule accepted 67 trials as known classes, of which 65 were correct. A nearest-trial baseline obtained 69/75 correct in a leave-one-trial-out evaluation. The class set was selected after preliminary analysis, so these results are internal, post-selection estimates. They do not measure generalisation to a new session, object instance, or unseen class.

Keywords: proprioception, LEAP Hand, household object recognition, physical grasping, neural network

Category: Robotics & Intelligent Machines

1. Introduction and Objectives

A robot can learn about an object through contact, even when a camera cannot see the contact surfaces. Earlier research has used joint positions and kinesthetic grasp information for object recognition (Sintov and Meir, 2023; Vásquez and Perdereau, 2017) but this project asks a narrower, practical question: are the final angles after a short, repeatable sequence of physical LEAP Hand grips enough to distinguish a set of household items? The objective is to build and evaluate a working physical-only recognizer that uses the hand's existing joint-position readings. A second objective is to

compare a compact neural classifier with a nearest-recorded-trial method. The study treats an empty-hand grasp as a class. It predicts a trained label, not the object's dimensions or un-trained object's identity.

2. Innovation

The experiment combines three deliberately different grips, finger-level stopping on motor-current thresholds, and a classifier that reads only the 48 final joint angles. This removes any need for vision or added tactile arrays during recognition, unlike most grasp based recognition found in robotic hands. The system also retains individual physical trials, supports retaking mistaken recordings, and reports uncertainty rather than forcing every measurement into a known class, it has a strict confidence rule, and retakes measurements until it is confident in the identified class. The innovation is an evaluated physical implementation of this specific sensing and decision pipeline.

3. Methodology

We used a four-finger, 16-actuator anthropomorphic dexterous LEAP Hand as created by Shaw et al. (2023). We programmed and ran with the same three configured grips for each item. Objects were placed near a consistent palm location. During each grip, all joints advanced toward preset targets. If a moving joint's measured motor current exceeded its threshold, the entire finger stopped, while other fingers continued. The controller then recorded all 16 final angles and moved to the next grip. Five physical trials were collected for each of 14 household items and for the empty-hand condition, yielding 75 real 48-angle vectors. Motor currents controlled stopping but were not classifier features.

Each angle was normalized using its configured start and target values. The neural model had 48 inputs, one 24-unit tanh hidden layer, and 15 softmax outputs. Training used Adam and small Gaussian angle perturbations, applied only to training folds. Three-fold validation kept real trials separate from their synthetic copies. The model's top class, confidence margin, and distance to recorded examples determined whether a reading was accepted, uncertain, or unknown. A nearest-trial baseline was evaluated by leaving out one real trial at a time. Saved trial CSV files and validation reports supported the analysis.

4. Results and Conclusions

The neural model classified 70 of 75 held-out physical trials correctly before the uncertainty rule, an internal accuracy of 93.3%. It accepted 67 readings as known; 65 of these were correct, while eight readings were uncertain. The separate nearest-trial evaluation obtained 69 correct labels out of 75. Most corresponding joint angles are close, while differences occur across multiple joints and grips; this motivates use of the complete angle vector rather than one hand-chosen threshold. The results establish feasibility under these recorded conditions, but independent sessions, new instances, and unseen classes still require prospective tests.

References

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